# OLYMPIC DAM, Australia

WMO: **956580**

Lat: **30.4828S** Lon: **136.8772E** Elev: **100** StdP: **100.13** Time Zone: **9.50 (AUA)** Period: **01-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **7** | **0.5** | **2.2** | **-13.5** | **1.2** | **27.1** | **-10.7** | **1.5** | **24.2** | **10.4** | **18.8** | **9.4** | **18.5** | **1.2** | **150** | **0.519** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **1** | **14.6** | **42.2** | **19.2** | **40.4** | **18.6** | **38.6** | **18.5** | **23.0** | **31.2** | **21.9** | **31.9** | **20.9** | **31.7** | **5.1** | **350** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **21.2** | **16.0** | **25.0** | **19.2** | **14.1** | **25.1** | **17.5** | **12.7** | **25.4** | **68.8** | **30.4** | **64.4** | **31.6** | **60.9** | **31.5** | **27.5** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **11.4** | **10.0** | **8.9** | DB | **-2.2** | **44.3** | **1.3** | **1.6** | **-3.2** | **45.4** | **-4.0** | **46.3** | **-4.7** | **47.2** | **-5.7** | **48.4** | (4) |
| (5) | | | | WB | **-3.5** | **24.5** | **1.6** | **1.8** | **-4.7** | **25.8** | **-5.6** | **26.9** | **-6.5** | **27.9** | **-7.7** | **29.2** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **20.1** | **29.0** | **27.7** | **24.7** | **20.4** | **15.4** | **11.8** | **11.5** | **13.0** | **16.8** | **20.7** | **24.1** | **26.6** | (6) |
| (7) | | | DBStd | **7.12** | **4.19** | **4.16** | **3.78** | **3.50** | **3.13** | **2.59** | **2.75** | **3.17** | **3.90** | **4.29** | **4.37** | **4.22** | (7) |
| (8) | | | HDD10.0 | **30** | **0** | **0** | **0** | **0** | **1** | **11** | **13** | **5** | **0** | **0** | **0** | **0** | (8) |
| (9) | | | HDD18.3 | **801** | **0** | **0** | **2** | **17** | **104** | **196** | **212** | **169** | **77** | **22** | **2** | **0** | (9) |
| (10) | | | CDD10.0 | **3716** | **589** | **496** | **456** | **313** | **166** | **66** | **60** | **99** | **204** | **331** | **422** | **513** | (10) |
| (11) | | | CDD18.3 | **1445** | **331** | **263** | **199** | **80** | **12** | **0** | **1** | **4** | **31** | **95** | **174** | **255** | (11) |
| (12) | | | CDH23.3 | **20468** | **5008** | **3771** | **2497** | **909** | **112** | **2** | **12** | **67** | **496** | **1393** | **2485** | **3716** | (12) |
| (13) | | | CDH26.7 | **11693** | **3216** | **2323** | **1313** | **351** | **19** | **0** | **1** | **12** | **183** | **678** | **1372** | **2226** | (13) |
| (14) | Wind | | WSAvg | **4.3** | **5.3** | **5.0** | **4.5** | **3.8** | **3.4** | **3.1** | **3.5** | **4.0** | **4.4** | **5.0** | **5.1** | **5.1** | (14) |
| (15) | Precipitation | | PrecAvg | **163** | **13** | **24** | **6** | **17** | **13** | **14** | **8** | **11** | **13** | **10** | **15** | **16** | (15) |
| (16) | | | PrecMax | **337** | **79** | **128** | **23** | **136** | **64** | **85** | **27** | **33** | **45** | **39** | **47** | **69** | (16) |
| (17) | | | PrecMin | **40** | **0** | **0** | **0** | **0** | **0** | **0** | **1** | **0** | **0** | **0** | **0** | **1** | (17) |
| (18) | | | PrecStd | **70** | **21** | **33** | **6** | **31** | **14** | **21** | **8** | **11** | **14** | **12** | **12** | **15** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **45.0** | **43.8** | **39.9** | **35.5** | **28.8** | **23.2** | **24.9** | **28.4** | **34.4** | **38.0** | **41.7** | **43.3** | (19) |
| (20) | | | MCWB | | **19.8** | **21.0** | **18.4** | **17.0** | **14.3** | **14.1** | **13.2** | **13.6** | **14.7** | **16.8** | **18.0** | **19.2** | (20) |
| (21) | | 2% | DB | | **42.5** | **41.4** | **37.6** | **32.8** | **26.3** | **20.5** | **22.2** | **25.0** | **31.2** | **35.5** | **38.8** | **41.0** | (21) |
| (22) | | | MCWB | | **19.4** | **19.5** | **17.7** | **16.3** | **14.3** | **12.3** | **12.0** | **11.8** | **13.9** | **15.9** | **17.6** | **18.4** | (22) |
| (23) | | 5% | DB | | **40.5** | **39.1** | **35.4** | **30.6** | **24.1** | **18.9** | **20.0** | **22.8** | **28.6** | **33.1** | **36.4** | **38.6** | (23) |
| (24) | | | MCWB | | **19.0** | **19.2** | **17.4** | **15.5** | **13.5** | **11.9** | **11.4** | **11.5** | **13.2** | **15.1** | **16.7** | **18.0** | (24) |
| (25) | | 10% | DB | | **38.6** | **37.2** | **33.3** | **28.5** | **22.0** | **17.5** | **18.2** | **20.7** | **26.0** | **30.5** | **33.9** | **36.3** | (25) |
| (26) | | | MCWB | | **18.7** | **18.7** | **17.3** | **15.2** | **12.8** | **11.2** | **10.5** | **10.9** | **12.4** | **14.5** | **16.2** | **17.5** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **24.6** | **25.1** | **23.5** | **20.6** | **16.8** | **15.8** | **15.2** | **15.5** | **16.9** | **18.8** | **21.6** | **23.7** | (27) |
| (28) | | | MCDB | | **30.3** | **28.0** | **27.6** | **22.8** | **22.8** | **19.8** | **20.2** | **22.9** | **24.8** | **30.3** | **29.8** | **29.6** | (28) |
| (29) | | 2% | WB | | **22.6** | **22.9** | **21.9** | **18.9** | **15.7** | **14.2** | **13.0** | **13.6** | **15.2** | **17.5** | **20.0** | **21.7** | (29) |
| (30) | | | MCDB | | **32.2** | **33.4** | **28.8** | **24.1** | **22.5** | **17.8** | **19.3** | **21.2** | **26.2** | **29.4** | **28.8** | **31.1** | (30) |
| (31) | | 5% | WB | | **21.6** | **21.6** | **20.4** | **17.5** | **14.7** | **13.1** | **12.0** | **12.5** | **14.2** | **16.5** | **18.9** | **20.5** | (31) |
| (32) | | | MCDB | | **33.3** | **32.9** | **29.6** | **26.4** | **21.2** | **17.0** | **18.8** | **20.7** | **25.2** | **29.0** | **29.5** | **31.3** | (32) |
| (33) | | 10% | WB | | **20.6** | **20.3** | **18.8** | **16.4** | **13.8** | **12.1** | **11.1** | **11.5** | **13.3** | **15.5** | **17.8** | **19.2** | (33) |
| (34) | | | MCDB | | **34.0** | **32.5** | **30.7** | **26.6** | **20.4** | **16.3** | **17.3** | **19.4** | **24.2** | **28.0** | **31.0** | **32.5** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **14.6** | **14.1** | **13.4** | **13.4** | **12.6** | **11.9** | **13.3** | **14.4** | **15.0** | **15.1** | **14.6** | **14.7** | (35) |
| (36) | | 5% DB | MCDBR | | **16.7** | **16.5** | **16.6** | **16.2** | **14.4** | **13.6** | **15.3** | **16.9** | **18.3** | **18.2** | **17.4** | **17.5** | (36) |
| (37) | | | MCWBR | | **5.3** | **5.2** | **6.1** | **6.4** | **6.5** | **7.5** | **8.0** | **8.1** | **7.4** | **7.1** | **6.1** | **5.9** | (37) |
| (38) | | 5% WB | MCDBR | | **12.5** | **12.6** | **10.7** | **11.2** | **11.4** | **10.5** | **13.7** | **14.7** | **15.3** | **14.4** | **13.1** | **12.9** | (38) |
| (39) | | | MCWBR | | **5.1** | **4.6** | **4.4** | **5.3** | **5.6** | **6.1** | **7.5** | **7.6** | **6.9** | **6.2** | **5.5** | **5.7** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.363** | **0.356** | **0.336** | **0.326** | **0.307** | **0.293** | **0.298** | **0.315** | **0.349** | **0.355** | **0.362** | **0.352** | (40) |
| (41) | | taud | | | **2.401** | **2.448** | **2.477** | **2.485** | **2.510** | **2.538** | **2.511** | **2.446** | **2.355** | **2.350** | **2.371** | **2.414** | (41) |
| (42) | | Ebn at Noon | | | **977** | **965** | **952** | **910** | **880** | **872** | **881** | **908** | **922** | **955** | **972** | **991** | (42) |
| (43) | | Edh at Noon | | | **127** | **117** | **107** | **95** | **83** | **76** | **82** | **97** | **119** | **128** | **130** | **126** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **8.07** | **7.35** | **6.32** | **4.89** | **3.71** | **3.16** | **3.46** | **4.46** | **5.83** | **6.92** | **7.62** | **8.09** | (44) |
| (45) | | RadStd | | | **0.40** | **0.44** | **0.39** | **0.21** | **0.21** | **0.18** | **0.13** | **0.17** | **0.29** | **0.37** | **0.36** | **0.32** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (3 neighbors) | **+0.40** | **N/A** | **N/A** | **+1.07** | **N/A** | **N/A** | **N/A** | **N/A** | **+158** | **+107** | (47) |

Nomenclature: See separate page